Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/30/2006
PI Name:	Vogel, Viola		
Project Title:	Motor Proteins as Shuttles for Directed Transport in Synthetic Matrices		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Other		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	98195	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	OTHER
Start Date:	01/01/1999	**End Date:**	01/01/2003
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			

Enabling technology for controlling transport processes is tightly linked to the size scale. Whereas sophisticated machines and devices have been developed for the macroscopic world, new principles have to be employed to control molecular motion on the nanoscale. Towards this end, nature has evolved specialized molecules, namely motor proteins, to actively transport molecules over long distances and against concentration gradients.
The long-term goal of this research is to use biological principles to create novel hybrid materials that are capable of dynamic spatial self-organization and modulation. Inspiration for this approach comes from cells which use motor proteins that move along polymeric tracks to actively transport cellular components to defined locations. Our specific goal is to build a molecular shuttle that is operated on a patterned array of motor proteins on nanoengineered surfaces. Objectives include the control of the direction of movement, and the ability to load and unload cargo at user-specified locations.

Task Description:	We view this as a first step toward the development of novel devices and materials in which chemically powered molecule-based active transport is utilized to control dynamic processes, including adaptive behavior, self-repair, spatially-confined self-assembly, and the modulation of optic and electronic properties. The long-term goal of this research is to use biological principles to create novel hybrid materials that are capable of dynamic spatial self-organization and modulation. The deliverable will be a toolbox of technologies to assemble functional transport systems on the nanoscale. Like a monorail at the nanoscale, the molecular shuttle system will be able to carry cargo, from molecules to microscopic devices, along engineered tracks. For each of the key areas of guiding, loading, and speed control, we develop solutions of how to control position, attachment of cargo and velocity of the molecular shuttles non-invasively. Molecular shuttles integrated into the next generation of hybrid materials could confer properties typically associated with living systems, including directed assembly/disassembly or self-repair, and find a range of applications in microfabricated devices.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)